

A real time reservoir inflow forecast evaluation framework

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Summary

Thank you for the opportunity to review this manuscript which presents a framework for real-time evaluation of reservoir inflow forecasts, demonstrated on the Great River Hydro system on the Connecticut River. The paper addresses a practical and important gap: how hydropower operators can systematically assess whether new forecast products justify their cost. The iterative approach with direct operator feedback is sensible and the tailored exceedance metric is a good example of translating generic forecast quality into operationally relevant information. I have two major comments regarding the need to engage with the forecast value literature given the paper's stated goal of informing investment decisions, and the positioning of the contribution. I also have some minor comments on clarity, completeness, and technical accuracy. The core work is solid and the practical contribution is genuine. Assuming the major comments are addressed I recommend this manuscript is accepted with minor revision.

Major comments

The stated goal of the framework is to help operators evaluate whether a forecast product will improve operations enough to justify investment. The paper does not include any assessment of forecast value and the authors acknowledge this gap (L172–177), framing it as “an important next step.” I appreciate this is a difficult problem, but the paper needs to at least engage with the existing forecast value literature and discuss how the presented metrics relate to the investment question. There is an established body of work on forecast value, from Thompson and Brier's (1955) first quantitative treatment through to more general decision-theoretic approaches such as Laugesen's (2023, 2026) Relative Utility Value method. The authors do not need to perform a full value assessment, but they should reference this literature, discuss how it connects to their framework, and provide operators with at least a conceptual pathway from the metrics presented to the investment decision the paper is motivated by.

The paper would benefit from clearly articulating what is new relative to existing frameworks. The five-step iterative process (collect data, compute metrics, visualise, solicit feedback, iterate) reads more as standard good operational practice than a novel framework contribution. I feel that the tailored metrics and operator feedback loop are where the genuine contribution lies, especially in the real-time context, and these deserve more prominence and deeper treatment.

Minor comments

L2 – Title: “real time” should (perhaps) be hyphenated as “real-time” when used as an adjective (also applies throughout the manuscript, e.g. L44, L48, L50, L110, L116).

L12–13 – The sentences seem to imply that we should expect verification methods to improve alongside improvements in forecasts, but this isn't the case. Also, the first sentence is about forecast and scheduling methods but the second sentence links it to operational paradigms somewhat abruptly. Suggest restructuring so the research gap is clearer.

L42–44 – The distinction between deterministic and probabilistic is important and well noted. However, the dismissal of ensemble evaluation metrics seems premature given that the authors later suggest probabilistic forecasts may help (L170). Consider briefly noting what ensemble metrics could be added if ensemble forecasts were available.

L45–47 – The argument against hindcast evaluation is reasonable but somewhat overstated. Hindcast evaluations do not necessarily require operators to re-operate the system, especially if verifying forecasts of environmental variables such as inflows, which you are. Suggest softening this claim or explaining how operators are in the loop for the inflow forecasts. The first major comment above may be relevant here.

L71 – Describing persistence as “arguably worst possible forecast” is incorrect. Persistence is a competitive benchmark for short lead times as the authors themselves note in the next sentence. Climatology is typically considered the lower bound for longer horizons. Suggest rephrasing to “a simple but competitive benchmark.”

L80–81 – A triangular smoother is reasonable but the choice of window width is not discussed. The smoothing window will affect the comparison with forecasts and should be selected with care. Please note the window used and perhaps the sensitivity to this choice on your findings.

L92 – “King-Gupta Efficiency” should be “Kling-Gupta Efficiency.” Also, “(refs)” is a placeholder that needs to be replaced with actual references (Gupta et al. 2009 presumably).

L92–93 – The claim that KGE is not skewed by large values while NSE and RMSE are requires clarification. KGE and NSE both use correlation as a component. The relevant difference is that KGE decomposes performance into correlation, variability ratio, and bias ratio rather than using squared error. The statement as written is misleading.

L95–96 – The KGE threshold of -0.41 is attributed to comparing against the mean of observations but the Knoben et al. (2019) citation is on L97, not adjacent to the claim. Consider restructuring so the citation directly supports the threshold statement.

L98–99 – “The bias of a single forecast point is the difference between observed and forecasted values” defines error (sometimes called residual), not bias. Bias is a systematic tendency. Suggest correcting the definition.

Eq. 3 – The Pearson correlation coefficient formula is missing a summation over the second term of denominator.

L107 – 589 megawatts of “nominal” hydropower capacity. Is this installed capacity or average generation? Please clarify.

L109 – Add a sentence describing how these 3 sites are related, given they are downstream from one another one could assume they are operated jointly. Also good to note why you chose these sites over all the other sites (green dots in the figure)

L117 – Anonymising the commercial forecast products is understandable but limits reproducibility. Consider at minimum describing the general class of model used by each (e.g., physics-based, statistical, machine learning) so readers can contextualise the results.

Figure 1 – Y-axis units are “cfs” (cubic feet per second). Consider also providing SI units for international readers.

L134 – “Forecasts tend to perform worse in the winter which is the wet season and worse in the summer which is the dry season.” This says forecasts perform worse in both wet and dry seasons. Please clarify when forecasts actually perform well and why. The following sentence about snow melt season partially addresses this but the logic is unclear.

L146 – The 0.5 foot (≈ 15 cm) operating range is an important constraint that motivates a strong tailored metric. However, the forebay exceedance analysis assumes an inflow=outflow condition which is operationally unrealistic as the authors note. It would strengthen the analysis to discuss how operators actually deviate from this assumption and what the real-world exceedance rates might be.

L178–182 – The discussion on limitations of real-time vs hindcast evaluation is good. Consider also noting the limitation that almost two years of data may not capture the full range of extreme events which are often the conditions where forecast quality matters most for operational decisions.

Recommendation

Minor revision. The practical application and operator engagement aspects are valuable, and the tailored metrics are a genuine contribution. The paper needs to be better positioning, particularly against the forecast value literature to connect the presented metrics to the investment decisions the paper is motivated by. These are framing and positioning issues rather than fundamental methodological concerns.